Compound interest

1 Describe how compound interest is earned.

2 $\$9000$ is invested for 3 years at a rate of 5% p.a. compounded annually.

a Complete the table:

b Calculate the total interest accumulated over 3 years.

c Calculate the value of the investment at the end of 3 years.

	Interest ($)	Balance ($)
After 0 years	0	9000
After 1 year		
After 2 years		
After 3 years		

3 $\$3000$ is invested at 4% p.a., compounded annually. The table below tracks the growth of the principal over three years.

	Value at start of time period	Value at end of time period	Interest earned
1st year	$\$3000$	A	B
2nd year	C	$\$3244.80$	D
3rd year	$\$3244.80$	$\$3374.59$	E

a Find the value of:

i A ii B iii C iv D

v E

b Find the total interest earned over the three years.

4 Assuming that a year has 365 days and 52 weeks, calculate the final value of the following investments:

a A $\$7510$ investment earns interest at 4.5% p.a. compounded annually over 6 years.

b A $\$3400$ investment earns interest at 3% p.a. compounded quarterly over 19 years.

c A $\$9090$ investment earns interest at 4.7% p.a. compounded semiannually over 11 years.

d A $\$8920$ investment earns interest at 3.3% p.a. compounded monthly over 5 years.

e A $\$3420$ investment earns interest at 4.9% p.a. compounded weekly over 9 years.

f A $\$6000$ investment earns interest at 5% p.a. compounded daily over 11 years.

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5 Joan's investment of $\$3000$ earns interest at a rate of 3% p.a, compounded annually over 4 years. What is the value of the investment at the end of the 4 years?

6 Calculate the amount, A, that an investment of $\$1000$ compounded annually is worth after:

a 3 years at an interest rate of 4% p.a. **b** 4 years at an interest rate of 9% p.a.

7 $\$4000$ is invested in a term deposit at a rate of 3% per quarter compounded monthly. Find the value of the investment after 8 years.

8 $\$3000$ is invested in a term deposit at a rate of 1% per month compounded quarterly. Find the value of the investment after 6 years.

9 Amelia borrows $\$2400$ at a rate of 6.3% p.a. compounded annually. If she pays off the loan in a lump sum at the end of 5 years, how much interest does she pay?

10 $\$380$ is invested at 2% p.a. compounded annually for 5 years. At the end of 5 years, the entire value of the investment is reinvested at 3% p.a. compounded annually for 4 more years. What is the final value of the investment at the end of the 9 years?

11 Kate invests $\$3000$ at a rate of 2% p.a. compounded annually. Find how much the investment is worth after:

a 24 months **b** 18 months **c** 30 months

12 Ben borrows $\$7000$ at a rate of 2% p.a. compounded annually. After 2 years he makes a repayment of $\$500$. After another 3 years, with no further repayments, how much does Ben owe?

13 Xavier invests $\$7000$ in a term deposit with a rate of 2% p.a. compounded annually. After 3 years he withdraws $\$600$, and leaves the rest in the the term deposit for 2 more years. How much is the investment worth after the total 5 years?

14 Assuming that a year has 52 weeks and 365 days, calculate how much is owed at the end of the following loan periods, given that no repayments are made:

a $\$8000$ is borrowed at a rate of 5.4% p.a. compounded semi-annually for 5 years.

b $\$4000$ is borrowed at a rate of 2.8% p.a. compounded quarterly for 3 years.

c $\$6000$ is borrowed at a rate of 20% p.a. compounded annually for 5 years.

d Emma borrows $\$7000$ is borrowed at a rate of 4.7% p.a. compounded annually for 3 years.

e $\$6000$ is borrowed from a bank at a rate of 7.2% p.a. compounded weekly for 5 years.

f $\$7000$ is borrowed from a bank at a rate of 3.4% p.a. compounded monthly for 6 years.

g $\$2000$ is borrowed from a bank at a rate of 1.3% p.a. compounded daily for 3 years.

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15 Lachlan borrows $\$5000$ at a rate of 4.5% compounded annually. After 2 years the bank increases the interest rate to 4.6%. If he pays off the loan in a lump sum at the end of 5 years, how much interest does he pay?

16 Sean borrows $\$7000$ at a rate of 5.5% p.a. compounded weekly. If she pays off the loan in a lump sum at the end of 5 years, find how much interest she pays. Assume there are 52 weeks in a year.

17 Assuming that in a year there are 52 weeks and 365 days, calculate the amount of interest earned on the following investments:

a Sally's investment of $\$4200$ earns interest at 2.7% p.a. compounded quarterly over 13 years.

b Sally's investment of $\$3070$ earns interest at 4.5% p.a. compounded monthly over 7 years.

c Han's investment of $\$9110$ earns interest at 3.2% p.a. compounded weekly over 18 years.

d Luke's investment of $\$6220$ earns interest at 2.8% p.a. compounded daily over 11 years.

e Buzz's investment of $\$4920$ earns interest at 5% p.a. compounded semiannually over 2 years.

f Sally's investment of $\$8910$ earns interest at 4% p.a. compounded annually over 10 years.

18 Pauline borrows $\$50\,000$ at a rate of 5.4% per annum. If she pays off the loan in a lump sum at the end of 7 years, find how much interest she pays if the interest is compounded:

a Daily **b** Monthly **c** Quarterly

19 Emma wants to invest $\$1400$ at 5% p.a for 5 years. She has two investment options, compounding quarterly or compounding monthly.

a Calculate the value of the investment if it is compounded quarterly.

b Calculate the value of the investment if it is compounded monthly.

c Calculate how much extra the investment is worth if it is compounded monthly rather than quarterly.

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20 Maria has $\$9000$ to invest for 5 years and would like to know which investment plan to enter into out of the following three.

- **Plan 1**: invest at 4.51% p.a. interest, compounded monthly
- **Plan 2**: invest at 6.16% p.a. interest, compounded quarterly
- **Plan 3**: invest at 5.50% p.a. interest, compounded annually

a Calculate the future value of the investment under Plan 1.

b Calculate the future value of the investment under Plan 2.

c Calculate the future value of the investment under Plan 3.

d Which investment plan yields the highest return?

21 Mae's investment into a 20-year 2.33% p.a. corporate bond grew to $\$13\,600$. Calculate the size of Mae's initial investment if interest was compounded:

a Annually

b Half-yearly

c Quarterly

d Monthly

e Weekly, assuming there are 52 weeks in a year.

f Daily, assuming there are 365 days in a year.

22 Katrina borrows $\$6500$ at a rate of 6.6% p.a. compounded semi-annually. If she pays off the loan in a lump sum at the end of 5 years, find how much interest she pays.

23 Frank is working out the compound interest accumulated on his loan. He writes down the following working:

$$A = 6000\left(1 + \frac{0.08}{4}\right)^{(7\times 4)}$$

a How much did he borrow in dollars?

b What is the annual interest rate as a percentage?

c Is the interest being compounded weekly, monthly, quarterly or annually?

d For how many years is he accumulating interest?

e How much interest does he pay?

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Spreadsheets

24 The following spreadsheet shows the balance in a savings account in 2011, where interest is compounded quarterly:

	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1	$2000	$20	$2020
3	2	$2020	$20.20	$2040.20
4	3	$2040.20	$20.40	$2060.60
5	4			

a Calculate the quarterly interest rate.

b Complete the table for quarter 4.

c Use a spreadsheet to find how many quarters after the beginning of 2011 the balance will be double the initial investment of $2000.

25 The following spreadsheet shows the balance in a savings account in 2011, where interest is compounded monthly:

	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	January	$1000	$20	$1020
3	February	$1020	$20.40	$1040.40
4	March	$1040.40	$20.81	$1061.21
5	April	$1061.21	$21.22	$1082.43
6	May			

a Use the numbers for January to calculate the monthly interest rate.

b Complete the table for the month of May.

c Use a spreadsheet to calculate the total amount of interest earned over the year.

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26 The following spreadsheet shows the balance in a savings account where interest is compounded quarterly:

	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1		$100	$5100
3	2	$5100		$5202.00
4	3	$5202.00	$104.04	
5	4	$5306.04	$106.12	$5412.16

a Calculate the quarterly interest rate, correct to two decimal places.

b Complete the table.

27 The following spreadsheet shows the balance in a savings account in 2013, where interest is compounded monthly:

	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	July	$3000	$30	$\$X$
3	August	$3030	$30.30	$3060.30
4	September	$3060.30	$\$Y$	$3090.90
5	October	$\$Z$	$30.91	$3121.81
6	November	$3121.21	$31.22	$3153.03

Complete the table.

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